

UPS Meets Patch Queues

Evidence-timeline prioritization under limited capacity, cadence, and compliance gravity

Sergey Gordeychik
CyberOK Research
gordey@cyberok.ru

January 2026

Abstract

Modern vulnerability management must prioritize remediation under extreme disclosure volume and limited patch capacity. We introduce **UPS (Urgent Patch Score)**, an evidence-driven phase model that treats vulnerability urgency as a timeline of verifiable signals rather than a static score. Using the CISA KEV 2025 cohort ($N=245$) as a high-precision exploited set and a capacity-constrained patch simulation, we show that UPS enables meaningful shift-left remediation without using KEV as a trigger: 35–53% of KEV-2025 items are patched by the KEV addition date. With moderate capacity (≥ 20 items/week), UPS achieves near-perfect 14-day KEV compliance, while tighter deadlines (7 days) are primarily limited by organizational cadence rather than raw capacity. Finally, we quantify the core operational trade-off: earlier admission policies increase workload inflation (an upper-bound overhead factor, defined as total items handled per KEV-2025 item handled; up to $\sim 5\text{--}8\times$ under Watch+), whereas stricter thresholds (Track+/Prepare+) preserve much of the speed benefit at $\sim 2.6\text{--}2.8\times$ overhead. These results suggest that evidence-timeline prioritization can provide a practical, tunable alternative to score-only approaches when capacity and governance constraints dominate outcomes.

1 Introduction (or: why “just patch faster” is not a plan)

Vulnerability management is now an exercise in adult mathematics: **finite capacity, infinite inbox**. Tens of thousands of vulnerabilities appear each year, while remediation throughput is bounded by human time, testing, change windows, and the occasional release freeze that arrives exactly when you need it least.

A scoring zoo exists to help: CVSS for technical severity, EPSS for exploitation probability, plus assorted vendor scores and decision trees. But most of these tools answer the wrong question. The operational question is not “how scary is this CVE,” but:

Given our capacity and cadence, what will we patch on time—and what will we miss?

That question is naturally a *queueing* question. And “queueing” is where score-only prioritization tends to become a comforting spreadsheet that cannot survive contact with a backlog.

This paper takes a deliberately unromantic approach:

1. Treat urgency as a **timeline of evidence**, not a scalar number.

2. Convert the timeline into **phase-based priorities**.
3. Run it through a **capacity-bounded patch queue** with a real calendar.
4. Measure outcomes against **external compliance clocks**, not internal optimism.

2 Related Work (and why scores keep disagreeing)

2.1 Conflicting scores are not a bug, they are a feature

Koscinski et al. show empirically (and outcome-linked) that different scoring systems are *not interchangeable* and can disagree sharply on what is “top priority.”[2] This is not shocking: systems optimize different objectives, use different data, and carry different institutional incentives. The surprise is that practitioners still expect a single score to behave like an oracle.

2.2 EPSS-only prioritization meets the patch queue

Prediction is useful; assumptions are dangerous. In *Prediction Meets Patch Queues*, we evaluate the operational pattern “patch EPSS $> \tau$ ” under capacity constraints and show systematic limits when judged against exploited ground truth as represented by KEV additions in 2025.[1] That work motivates the next step: if score-only gates fail under queue dynamics, perhaps we should stop forcing a scalar to do the job of a timeline.

3 UPS: urgency as a timeline of verifiable signals

UPS (Urgent Patch Score) is not a “better number.” It is a **phase machine** driven by timestamped evidence.

3.1 Phases (human-readable, operationally unpleasant)

UPS models a vulnerability’s lifecycle through phases such as:

- **Radar**: “exists” but not yet operationally actionable.
- **Watch**: early public signals justify lightweight attention.
- **Track**: stronger evidence suggests rising exploitation likelihood.
- **Prepare**: practical readiness work should be underway.
- **Urgent Patch / Emergency**: patch now, argue later.

The labels map to actions (triage depth, detection effort, change planning), not just risk vocabulary.

3.2 Signals (vendor-neutral, but market-supplied)

UPS consumes public artifacts as evidence events: (*type, timestamp, evidence*). Examples include advisories, PoCs, exploit modules/templates, detection rules, exploitation reporting, and exploited catalogs such as CISA KEV.[7] A key design choice is that UPS does not require *all* signals to exist. It only requires that *some* signals exist early enough, often enough, to move work left.

Table 1: Priority stability mechanisms in UPS (the anti-flapping kit).

Mechanism	Operational effect
Sticky stage (monotone)	Prevents “today urgent, tomorrow meh” oscillation; once you pay attention, you keep the receipt.
Floor stage (high-precision evidence)	High-confidence signals (e.g., exploited-catalog additions) set a non-negotiable minimum.
Policy floor (admission threshold)	Tunable noise filter: Watch+/Track+/Prepare+ define how early you are willing to spend effort.
Event auditability	Each transition is tied to a timestamped artifact; post-mortems become boring (a compliment).

3.3 UPS phase machine and stability rules

UPS is deliberately conservative about priority oscillation. Two principles dominate: (i) evidence can escalate urgency, and (ii) urgency should not de-escalate without explicit policy.

Priority stability mechanisms. Table 1 summarizes the rules that prevent oscillation and make “why was this urgent?” answerable.

State update rule. Given a stream of events sorted by timestamp, UPS computes an instantaneous stage suggestion (`stage_point`), a non-decreasing floor stage derived from high-precision evidence (`floor_stage`), a policy minimum stage (`policy_min`), and a monotone aggregate stage (`stage_sticky`). The result is a phase timeline suitable for priority scheduling.

Listing 1: UPS phase update (pseudocode).

```
# Inputs:
# events: list of evidence events, each with (ts, type, fields)
# policy_min: admission threshold (Watch+/Track+/Prepare+)
# map_stage_point(type): maps event type -> instantaneous suggested stage
# map_floor(type): maps high-precision types -> floor stage (or None)

stage_sticky = RADAR # evidence-derived, monotone
floor_stage = RADAR # evidence-derived, monotone
admitted = False
admit_ts = None

for e in sort_by_ts(events):
    # 1) instantaneous suggestion from THIS event only
    stage_point = map_stage_point(e.type)

    # 2) update high-precision floor (if applicable)
    f = map_floor(e.type)
    if f is not None:
        floor_stage = max(floor_stage, f)

    # 3) evidence-derived urgency (monotone, independent of policy)
    stage_sticky = max(stage_sticky, stage_point, floor_stage)

    # 4) policy gates admission; it does not rewrite evidence state
    if (not admitted) and stage_sticky >= policy_min:
        admitted = True
```

```

    admit_ts = e.ts # optionally: next_planning_tick(e.ts)

# optional: append raw evidence artifacts to an audit ledger (independent of
    admission)
# record_evidence(e)

# emit phase timeline only once the item is admitted (policy-gated)
if admitted:
    emit(ts=e.ts,
        stage_point=stage_point,
        floor_stage=floor_stage,
        stage_sticky=stage_sticky,
        admitted=admitted,
        admit_ts=admit_ts)

```

4 Data

4.1 UPS timelines dataset

Our UPS signal collection yields **6,312 unique CVEs** (1999–2026) with at least one recorded event in `timelines.csv`. Of these, **5,492** enter an actionable (non-Radar) phase at least once.

For the year 2025, UPS highlights a small actionable subset:

- Watch+: 752 CVE-2025 items
- Track+: 536 CVE-2025 items
- Prepare+: 500 CVE-2025 items

This reflects the core premise: most CVEs do not receive strong external evidence signals and should not dominate scarce triage/patch capacity.

Release. Artifacts for this manuscript (including `timelines.csv`, figure-generation scripts, and the UPS signal taxonomy) are published at https://github.com/scadastrangelove/kev_vs_epss.

4.2 CISA KEV 2025 as high-precision exploited set

We use the CISA Known Exploited Vulnerabilities (KEV) catalog additions in 2025 ($N=245$) as a high-precision exploited cohort.[7] KEV is not perfect ground truth, but it is one of the clearest public “exploited means exploited” labels available.

We use *future-KEV* as a time-relative label: for a KEV-2025 CVE with KEV addition time t_{KEV} , at any earlier time $t < t_{\text{KEV}}$ the item is *future-KEV* (known by open signals but not yet KEV-listed). Shift-left is work completed by t_{KEV} without treating KEV itself as a trigger.

4.3 vKEV as a sensitivity check

We additionally analyze VulnCheck KEV (“vKEV”) as an alternative exploited-catalog timeline.[10] It is not a federal compliance standard in the way CISA KEV is, but it is useful for a robustness check on lead time and on “pre-catalog” evidence readiness.

5 Methods: patch queues, capacity, cadence, and the tyranny of calendars

5.1 Queue model (minimal assumptions, maximal honesty)

We simulate a discrete-time patch process:

- weekly cadence (planning tick),
- service capacity of 10/20/40 items per week,
- admission policies: Watch+, Track+, Prepare+ (policy minimum stage),
- scheduling by phase priority with escalation (Emergency > Urgent > Prepare > Track > Watch),
- constant service time per work item.

Yes, service time is constant. No, reality is not. That is the point: we isolate what public signals + bounded capacity can explain before adding organization-specific patch bundling, dependencies, and exposure probabilities.

5.2 Absolute SLA vs relative SLA (the part everyone confuses)

We report two SLA semantics because real life contains two clocks. **Absolute SLA** is calendar-anchored: deadlines are measured from an outside timestamp (here, KEV `dateAdded`) and do not care about internal ticket creation rituals.[8] Throughout, we use fixed 7-day and 14-day deadlines as *hypothetical internal SLAs* for comparability; they should not be read as due-date compliance, since catalog due dates can vary across entries. **Relative SLA** is the internal process view: deadlines are measured from discovery / scan time / ticket creation—a common framing in audit-driven workflows (e.g., patching critical vulnerabilities within 30 days in PCI DSS language).[9] Weekly cadence makes these diverge; this is not philosophy, it is arithmetic.

5.3 Cadence penalty and the “hotfix lane” upper bound

Weekly cadence introduces delay between an external event and the next planning tick. We quantify this as *cadence penalty* (0–6 days). We also compute a bounded “hotfix lane” upper bound: if urgent items can preempt mid-cycle, how much of strict absolute SLA failure is cadence-limited rather than capacity-limited?

6 Results

We evaluate nine scenarios (capacity $\times$ policy) on the KEV-2025 cohort. The figures below are intentionally repetitive: if a result cannot survive being plotted twice, it was never a result.

6.1 Shift-left: patching future-KEV before KEV arrives

UPS enables meaningful shift-left remediation without using KEV as a trigger: 35–53% of KEV-2025 items are already patched by the KEV addition date (day0), depending on policy and capacity (Figure S1; completion-time distribution in Figure S2). Interpretation: the system does not need KEV to start moving. It needs enough early signals to justify admission and preparation for a subset that later becomes KEV.

Table 2: KEV share and workload inflation in 2025 (upper bound; KEV is incomplete).

Cap/wk	Policy	Admitted	KEV share in admitted (%)	Overhead factor	Patched	KEV share in patched (%)	Overhead factor
10	Prepare+	520	39.2	2.55×	481	41.8	2.39×
10	Track+	551	36.7	2.73×	482	41.1	2.43×
10	Watch+	1388	13.0	7.67×	520	35.6	2.81×
20	Prepare+	520	39.2	2.55×	520	39.2	2.55×
20	Track+	551	36.7	2.73×	551	36.7	2.73×
20	Watch+	1388	13.0	7.67×	989	18.3	5.46×
40	Prepare+	520	39.2	2.55×	520	39.2	2.55×
40	Track+	551	36.7	2.73×	551	36.7	2.73×
40	Watch+	1388	13.0	7.67×	1366	13.3	7.55×

6.2 Signal latency vs ops latency: diagnose before prescribing

If you are late, first decide whether you were late to know (signal lag) or late to act (ops lag). Figure S3 makes this separation explicit.

6.3 7-day absolute compliance is often a cadence problem

For tight deadlines (7 days), cadence becomes the dominant limiter (Figure S4). A hotfix lane is not heroism; it is a calendar workaround with a budget.

6.4 ADSR-like completion envelope (what it is, and why it exists here)

We visualize patch completions relative to KEV `dateAdded` using an ADSR-like envelope—borrowed terminology from audio synthesis (Attack/Decay/Sustain/Release). It is a descriptive analogy: a way to show whether remediation behaves like a sharp “key press” spike at KEV, or whether preparation spreads the load earlier and softens the peak.

No, this is not a scientific claim that vulnerabilities are music. But if you have ever watched a backlog after a high-confidence exploited signal, the waveform comparison becomes... uncomfortably intuitive.

(*“Everything is music” remains a joke, not a theorem.*) UPS policies shift work left: visible pre-KEV activity reduces dependence on a single exploited-catalog trigger (Figure S5). The remaining post-KEV peak is the residual you did not prepare for (or could not safely patch early).

7 Workload inflation and the price of early admission

Shift-left remediation is not free. If KEV is treated as exploited ground truth, then work performed on non-KEV items looks like overhead. KEV is incomplete, so this is an upper bound; nevertheless, it is the operational budget you will be asked to justify.

Definition (upper-bound overhead). Let $\mathcal{W}$ be the set of work items admitted (or patched) under a scenario in 2025, and let $\mathcal{K} \subseteq \mathcal{W}$ be those items that are in KEV-2025. We report the *KEV share* $|\mathcal{K}|/|\mathcal{W}|$ and its inverse as an upper-bound *workload inflation/overhead factor*: $|\mathcal{W}|/|\mathcal{K}|$ (i.e., total items handled per KEV-2025 item handled). This is intentionally *not* normalized by the full KEV-2025 cohort size (245), because scenarios may fail to admit/patch a fraction of KEV items by the measurement time.

Supplementary Figure S6 shows the non-KEV share of admitted and patched work under each scenario (an upper bound on overhead because KEV is incomplete).

Table 3: Scenario summary (operating points).

Cap/wk	Policy	Admission overhead factor	Abs SLA@14d (%)	Hotfix pressure @7d (%)
10	Prepare+	2.55×	97.1	34.3
10	Track+	2.73×	97.6	33.9
10	Watch+	7.67×	98.0	30.6
20	Prepare+	2.55×	100.0	33.9
20	Track+	2.73×	100.0	32.2
20	Watch+	7.67×	100.0	29.0
40	Prepare+	2.55×	100.0	33.9
40	Track+	2.73×	100.0	32.2
40	Watch+	7.67×	100.0	28.2

8 Discussion: Why open-source intel is surprisingly effective

Security people like to complain that “everything is public, therefore everything is worse.” And yet most practical vulnerability intelligence pipelines—including ours—are fed primarily by public artifacts: advisories, PoCs, write-ups, detection rules, exploited catalogs, and the occasional dramatic blog post. The paradox is that **open-source intel is not merely available; it is often operationally sufficient** to drive prioritization under capacity constraints.

8.1 The market is a signal factory (and it can’t help itself)

A large fraction of “intel” exists because the security market has a chronic case of **information asymmetry**: buyers cannot directly observe competence, product quality, or response maturity. In such markets, actors emit **signals**—observable actions that credibly indicate hidden quality—because reputation and revenue depend on it. In Spence’s classic signaling model, signals are valuable precisely when quality is hard to observe and incentives exist to separate high- from low-quality players.[3]

In vulnerability practice, “signals” look like detailed advisories, technical root-cause analyses, exploitability notes, mitigations, detection content, and sometimes PoCs. These artifacts are not published out of kindness. They are published because **vendors must demonstrate competence, responsiveness, and relevance**—and the cheapest proof is public proof.

8.2 Disclosure is not a moral choice; it’s an equilibrium with externalities

Vulnerability disclosure is an ecosystem with multiple stakeholders (vendors, researchers, coordinators, brokers, regulators) and competing incentives; public information is one of the few coordination tools available. ENISA’s review frames disclosure as a set of incentive trade-offs and constraints rather than a single “responsible vs irresponsible” axis.[6]

The practical outcome is predictable: even when actors dislike disclosure, the ecosystem still tends to produce public artifacts because reputational and competitive pressure rewards visible work, and coordination failures are costly.

8.3 Information sharing can be individually rational (even for competitors)

If you squint hard enough, threat/vulnerability intel sharing is just another strategic interaction. Economic models show conditions where information sharing can be incentive-compatible and can reduce the tendency to defer security investments.[4, 5] In other words, some defensive artifacts are functionally information-sharing events—even when packaged as marketing.

8.4 Tooling amplifies weak signals into useful signals

Open tooling acts as a signal amplifier. One public artifact quickly becomes many artifacts: PoC → templates → detections → exploit discussions → “exploitation suspected” → exploited catalog. The original evidence may be incomplete, but the amplification chain produces enough structure for a prioritization system to operate.

This is where a timeline model like UPS benefits: the goal is not perfect truth; the goal is **early, verifiable timestamps for state transitions**. If the market reliably produces *some* early signals for a subset of vulnerabilities, we can shift work left, smooth peaks, and improve SLA outcomes under bounded capacity—without pretending we have omniscience.

8.5 The cynical takeaway (written politely)

Open-source intel is effective because the industry is forced to **perform competence in public**. The performance is not always honest, not always complete, and occasionally theatrical—but it is frequent, timestamped, and cross-referenced enough to be operationally valuable. UPS is designed to exploit exactly this property: treat public artifacts as evidence events, not absolute truth, and let queue dynamics reveal whether “early preparation” is worth the workload inflation.

8.6 Caveats: OSINT is biased, noisy, and strategically curated

We should not romanticize this. Market-driven signals come with predictable pathologies: marketing bias, attention bias, and adversarial noise. This is why UPS treats signals as inputs to a phase machine rather than as a scalar truth score, and why policy thresholds (Track+/Prepare+ vs Watch+) exist as tunable defenses against noise-driven workload inflation.

8.7 Research directions (without new assumptions)

A clean next step is to formalize the “market signal” hypothesis empirically:

- **Signal production elasticity:** how disclosure volume/quality responds to competitive intensity.
- **Signal-to-queue value:** how earlier public signals reduce time-to-admission and peak backlog under realistic cadences.
- **Cross-catalog lead time:** how alternative exploited catalogs behave as earlier/later public signals and what that implies for shift-left policies.

9 vKEV as a sensitivity check

We treat vKEV (VulnCheck’s exploited catalog) as a robustness check, not as a compliance clock. The point is not to crown a winner; it is to ask whether the UPS evidence stream carries meaningful signal *before* an exploited-catalog label arrives, and how exploited-catalog timelines compare.

9.1 Pre-vKEV evidence readiness (excluding vKEV as a signal)

For vKEV additions in 2025 ($N=888$), we compute the UPS phase immediately *before* the vKEV listing timestamp while excluding the VKEV_ADDED event from the evidence stream. This is the “no answer key” check: how far the pipeline progresses on open signals alone. Stage distribution is shown in Supplementary Figure S7.

Table 4: Pre-vKEV admission rates in 2025 while excluding VKEV_ADDED as an evidence signal ($N=888$). “Weekly tick” applies the next planning boundary after the pre-vKEV evidence time.

Threshold	Pre-vKEV evidence	%	Admitted by weekly tick	%
Watch+	399	44.9	386	43.5
Track+	181	20.4	167	18.8
Prepare+	172	19.4	159	17.9

9.2 Pre-admission rates under Watch+/Track+/Prepare+

We also measure whether each vKEV-added item would have been admitted *before* vKEV under the three admission thresholds, both in calendar time and after applying a weekly planning tick. The headline is that open signals provide non-trivial early readiness for a large minority under Watch+, and a smaller but meaningful fraction under Track+/Prepare+.

9.3 Lead time of exploited catalogs: vKEV vs CISA KEV

For CVEs appearing in both catalogs in 2025 ($N=196$ overlap), we compute the distribution of (CISA KEV `dateAdded` – vKEV `dateAdded`). Positive values mean vKEV is earlier. Median lead is modest (~ 2 days), but the tail is wide—the market does not synchronize its bragging rights (Supplementary Figure S8).

10 Limitations

- **No asset context (universal presence).** We simulate queue dynamics conditional on a CVE being in-scope; we do not model whether a specific organization is actually exposed to each CVE. Therefore absolute admitted/patched volumes are upper bounds; in deployment, inventory/exposure filtering shrinks both workload and the set of KEV items in scope, and shift-left should be measured on the in-scope subset.
- **Constant service time.** Patch effort is heterogeneous; bundling exists; dependencies exist. We model one unit per item to isolate signal + capacity effects.
- **Catalog bias.** KEV is incomplete ground truth; “overhead vs KEV” is an upper bound.
- **Cadence is simplified.** Weekly batching is common, but real organizations have mixed cadences and exception workflows.

11 Conclusion

UPS reframes vulnerability urgency as an evidence timeline and evaluates it where it matters: a capacity-bounded queue subject to compliance clocks. On KEV-2025, UPS enables meaningful shift-left remediation without using KEV as a trigger, reaches near-perfect 14-day absolute compliance at moderate capacity, and makes the speed-versus-workload trade-off explicit. If scores compete in spreadsheets, queues decide in production.

Key findings.

- UPS delivers shift-left remediation without treating KEV as a trigger (future-KEV work is measurable, not rhetorical).
- For 14-day external deadlines, moderate capacity is sufficient; for 7-day deadlines, cadence dominates and motivates a controlled hotfix lane.
- Earlier admission increases workload inflation; Track+/Prepare+ retain most of the speed benefit at far lower overhead than Watch+.

References

- [1] Sergey Gordeychik. *Prediction Meets Patch Queues: Empirical Limits of EPSS-Only Prioritization Using CISA KEV Additions in 2025*. TechRxiv preprint, 2026. DOI: 10.36227/techrxiv.176857939.95987957/v1.
- [2] Viktoria Koscinski, Mark Nelson, Ahmet Okutan, Robert Falso, and Mehdi Mirakhorli. *Conflicting Scores, Confusing Signals: An Empirical Study of Vulnerability Scoring Systems*. arXiv:2508.13644, 2025.
- [3] Michael Spence. Job market signaling. *The Quarterly Journal of Economics*, 87(3):355–374, 1973. DOI: 10.2307/1882010.
- [4] Esther Gal-Or and Anindya Ghose. The economic incentives for sharing security information. *Information Systems Research*, 16(2):186–208, 2005. DOI: 10.1287/isre.1050.0053.
- [5] Lawrence A. Gordon, Martin P. Loeb, William Lucyshyn, and Lei Zhou. The impact of information sharing on cybersecurity underinvestment: A real options perspective. *Journal of Accounting and Public Policy*, 34(5):509–519, 2015. DOI: 10.1016/j.jaccpubpol.2015.05.001.
- [6] ENISA. *Economics of Vulnerability Disclosure*. European Union Agency for Cybersecurity, December 2018.
- [7] CISA. Known exploited vulnerabilities catalog. U.S. Cybersecurity and Infrastructure Security Agency.
- [8] CISA. Binding Operational Directive 22-01: Reducing the Significant Risk of Known Exploited Vulnerabilities. U.S. Cybersecurity and Infrastructure Security Agency, November 2021.
- [9] PCI Security Standards Council. *Payment Card Industry Data Security Standard (PCI DSS) v4.0.1*. June 2024.
- [10] VulnCheck. VulnCheck KEV (Known Exploited Vulnerabilities): Documentation. VulnCheck Community Docs.

Supplementary figures

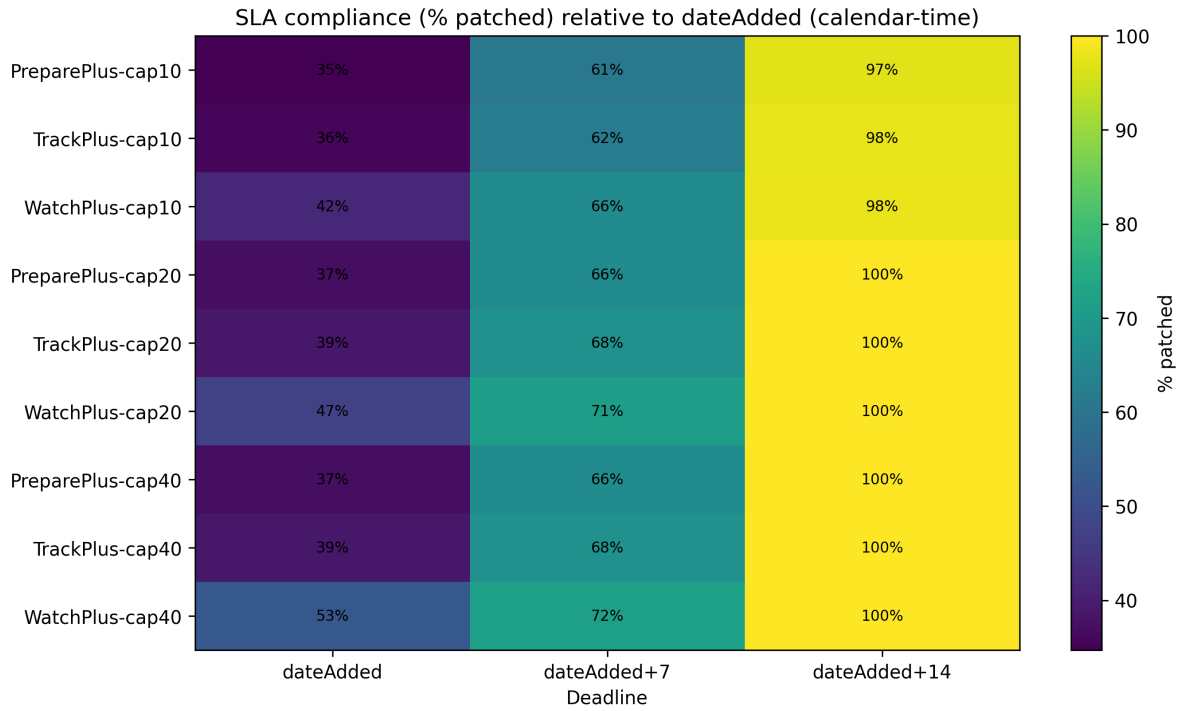


Figure S1: Pre-KEV patching rates (KEV-2025). Heatmap of the fraction of KEV-2025 items patched by (i) the KEV addition date (negative/zero lateness), (ii) within 7 days, and (iii) within 14 days, under nine scenarios (capacity $\times$ admission policy).

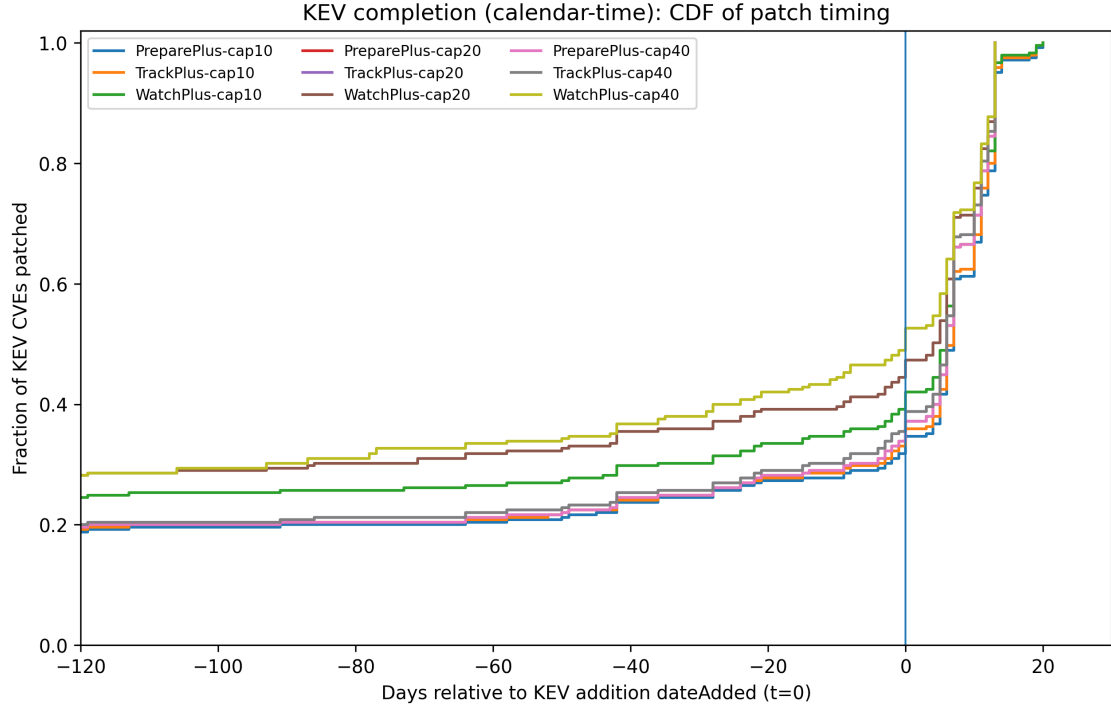


Figure S2: Completion ECDF relative to KEV `dateAdded`. Empirical CDF of completion time offsets for KEV-2025 across scenarios; values below 0 represent shift-left remediation.

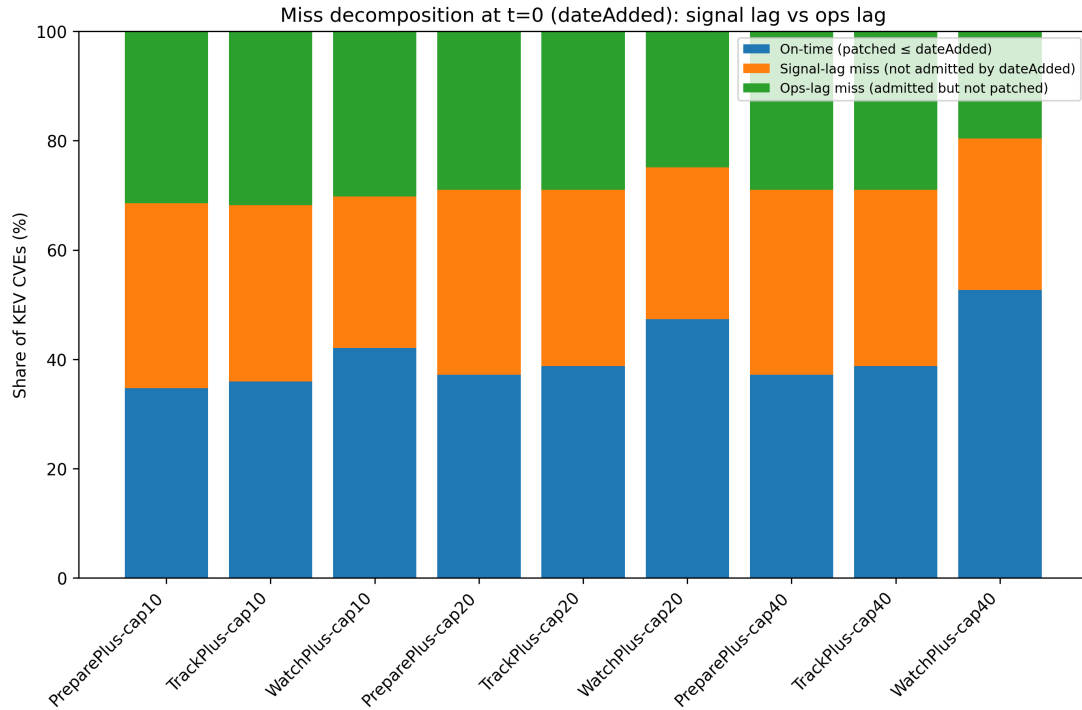


Figure S3: Miss decomposition: signal latency vs ops latency. For KEV-2025, we separate lateness into (i) evidence/signal lag (“late to know”) and (ii) operational/cadence lag (“late to act”).

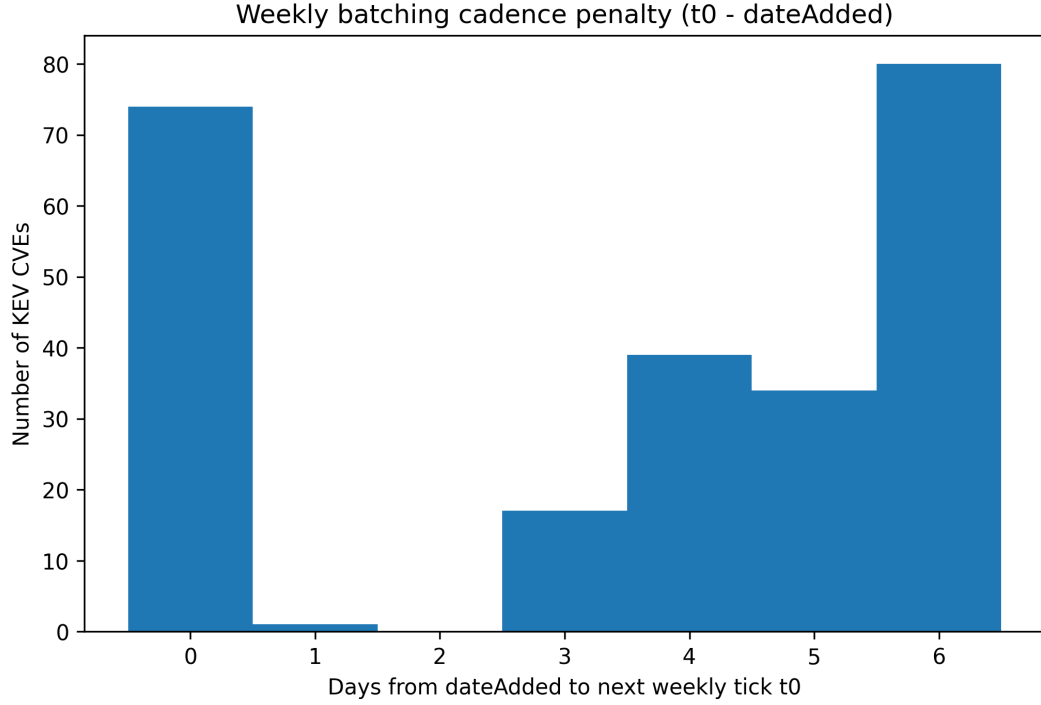


Figure S4: Cadence penalty and hotfix lane upper bound. The weekly planning tick introduces a 0–6 day delay between an external event and actionable admission; a bounded hotfix lane estimates how much of 7-day absolute SLA failure is cadence-limited rather than capacity-limited.

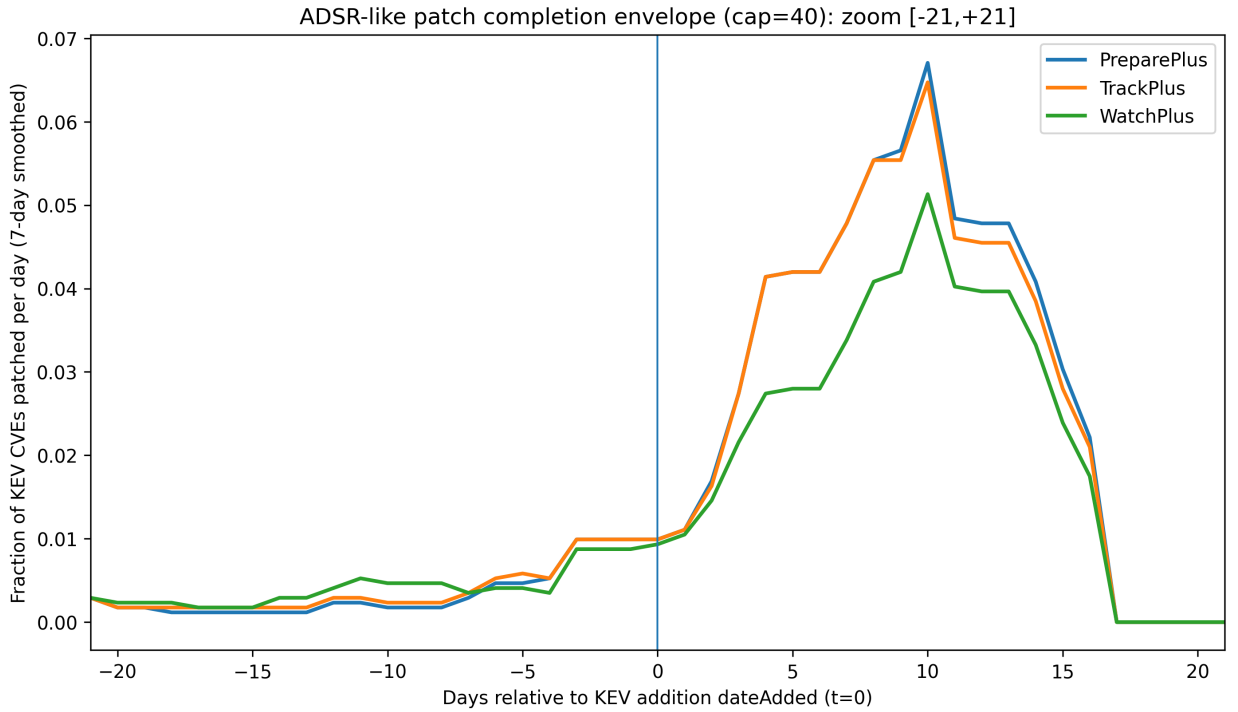


Figure S5: ADSR-like completion envelope around KEV `dateAdded`. Patch completion density shown as an “attack/decay/sustain/release”-style envelope to illustrate whether the remediation waveform spikes at KEV or is spread earlier by preparation.

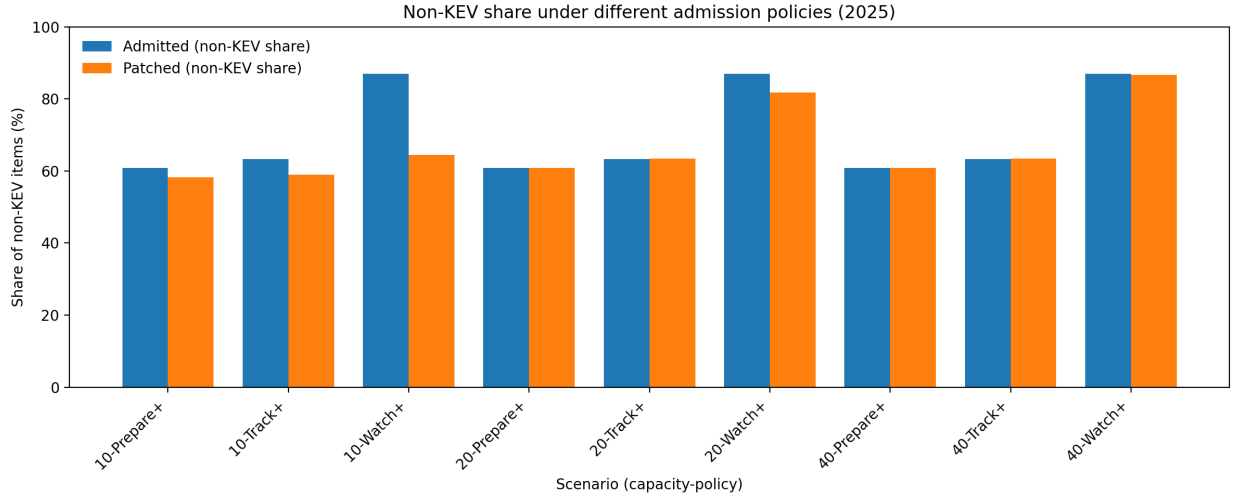


Figure S6: Non-KEV share of work (upper bound). Share of admitted and patched items that are not in KEV-2025, by scenario. Since KEV is incomplete, this should be interpreted as an upper bound on “overhead”.

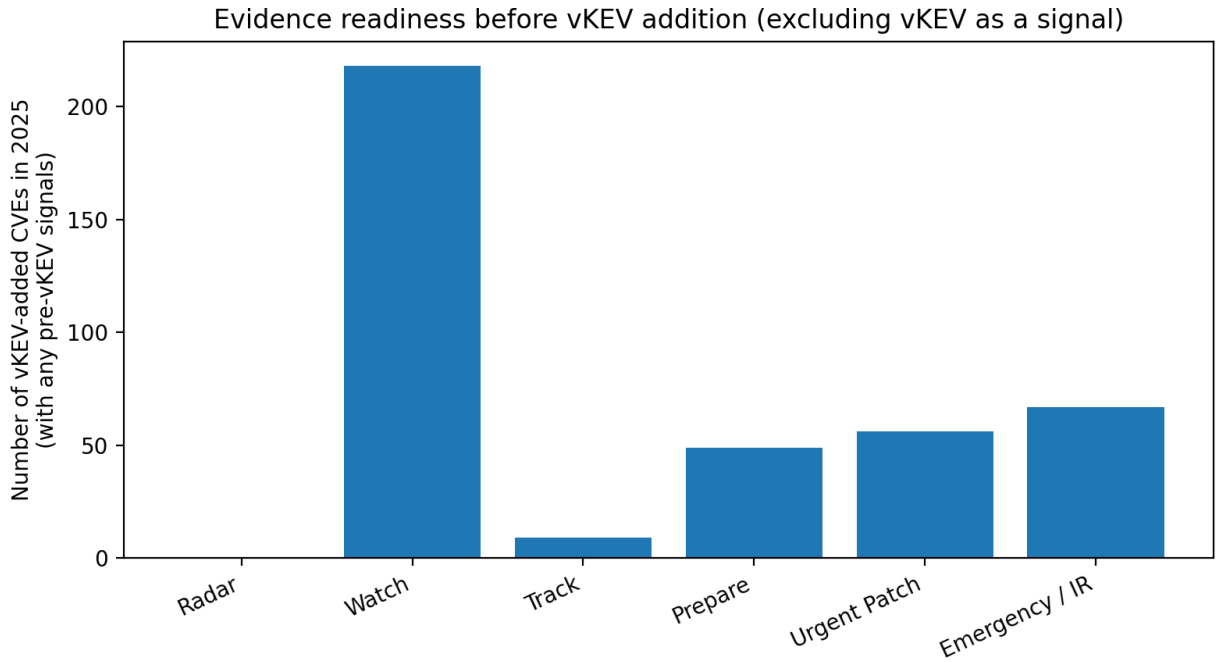


Figure S7: Pre-vKEV evidence readiness without “using the answer key”. For vKEV additions in 2025 ($N=888$), we compute the UPS phase immediately before the vKEV listing timestamp while excluding the `VKEV_ADDED` event from the evidence stream.

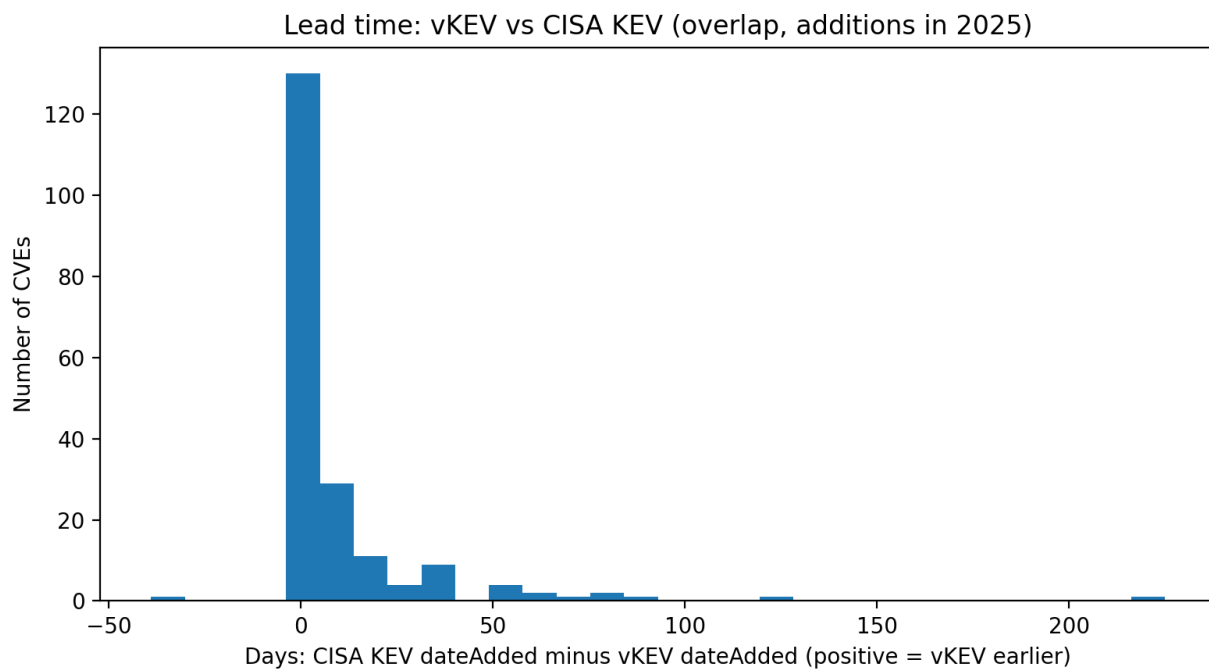


Figure S8: Lead time of alternative exploited catalogs. Distribution of (CISA KEV `dateAdded` – vKEV `dateAdded`) for CVEs appearing in both catalogs in 2025 ($N=196$). Positive values mean vKEV is earlier.